Modbus TCP/IP

Implementation guideline

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Introduction

- Supported on all products
- Implemented from software version 2.10.0.x

- Purposes
  - Read out of data measurements and device status information
  - Setpoints for Energy Managment Systems

Configuration parameters

- *modbus.tcp.enabled*
  - Enable/disable ModBus over TCP. This will enable the ModBus TCP server on the charging station. A default set of ModBus registers will be available to read out currents, voltages, states, etc. Consult the ModBus API documentations for more information.
  - Default value = true

Configuration parameters

- *modbus.tcp.port*
  - Unsecured ModBus TCP port
  - Default value = 502

Configuration parameters

- *modbus.tcp.ems.enabled*
  - Enable/disable EMS over ModBus TCP. This enables EMS mode (Energy Management System) on the charging station. Additional registers will become available to control the charging station like setting the current offered, reading additional transaction information, etc. Note: modbus.tcp.enabled is required to be true to enable EMS over ModBus TCP. Consult the ModBus API documentations for more information.
    Note: Only applicable for, and supported on, charging station type “Stand-Alone”
  - Default value = false

Register mapping

Holding Registers (4xxxx)

Description	Address	Register	# Registers	Type	Read/Write	Additional Information
API Version	0	1	2	int16	R	/
Number of phases	50	51	1	int16	R	Number of connector output phases
Max Amp per phase	51	52	1	int16	R	/
OCPP status	52	53	1	int16	R	0=Offline, 1=Online
Load shedding	53	54	1	int16	R	0=Disabled, 1=Enabled
Lock state	54	55	1	int16	R	0=Unlocked, 1=Locked
Contactor state	55	56	1	int16	R	0=Disengaged, 1=Engaged
LED index	56	57	1	int16	R	/
Measured Current L1	200	201	1	int16	R	mA
Measured Current L2	201	202	1	int16	R	mA
Measured Current L3	202	203	1	int16	R	mA
Measured Voltage L1	203	204	1	int16	R	V
Measured Voltage L2	204	205	1	int16	R	V
Measured Voltage L3	205	206	1	int16	R	V
Power Active Total	206	207	1	int16	R	W
Power Active L1	207	208	1	int16	R	W
Power Active L2	208	209	1	int16	R	W
Power Active L3	209	210	1	int16	R	W
Installation Current L1	210	211	1	int16	R	mA
Installation Current L2	211	212	1	int16	R	mA
Installation Current L3	212	213	1	int16	R	mA
Active Energy Import total	213	214	2	int32	R	Wh
Mode 3 State Numeric	300	301	1	int16	R	See mode 3 table →
Mode 3 State	301	302	2	string	R	See mode 3 table →
PWM (Ampere)	303	304	1	int16	R	mA
PWM	304	305	1	int16	R	/
PP	305	306	1	int16	R	A
CP+	306	307	1	int16	R	V
CP-	307	308	1	int16	R	V
Current Offered	400	401	1	int16	R/W	mA (Only writable when EMS setting is available)
Transaction Token	401	402	16	string	R	Only readable when EMS setting is enabled

Mode 3 State

Numeric	String
0	A1
1	A2
2	B1
3	B2
4	C1
5	C2
6	D1
7	D2
8	E
9	F

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